Manav Dheep Chandaka

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EDUCATION

University of Illinois at Urbana-Champaign *Urbana, IL* | Expected graduation: December 2027
Bachelor of Science in Mechanical Engineering | Minor in Computer Science

Relevant Coursework: *Statics, Dynamics, Solid Mechanics, Fluid Dynamics, Computer-Aided Design, Design for Manufacturability, Mechanical Design, Mechanical Systems, Engineering Materials, Discrete Structures, Data Structures, Electronic Circuits*

EXPERIENCE

Novel Mobile Robots Lab (NMbL) | *Undergraduate Researcher* *Urbana, IL* | August 2025 - Present

- Developing a suction-based climbing robot that integrates the GelBelt tactile sensing module, in collaboration with RoboTouch Lab, to achieve autonomous mapping and surface reconstruction for inspection tasks (Portfolio)
- Experimentally modeled and validated suspension and adhesion systems, analyzing force distribution and robot geometry to qualify stability and crawl to achieve 0.97 surface reconstruction accuracy on diverse structures such as aircraft skins
- Designed chain-drive, suspension, magnetic valve, vacuuming, guide rail, electronic integration, and active suction to control attachment and detachment

SIGRobotics | *Executive Chair* *Urbana, IL* | September 2024 - Present

- Led CAD design (Fusion 360), fabrication, and development of a 3-wheel holonomic-drive mobile manipulator; collaborated with Hugging Face LeRobot and ROBOTIS on open-source integration - 1450+ stars and 100+ community builds (GitHub)
- Worked with SO-100 robot arms, K-Scale humanoids, and TurtleBots on teleoperation, data collection, and autonomous control projects; handled mechanical assembly, electrical infrastructure, and multi-modal testing and deployment
- Simulated robotic systems in MuJoCo and performed hardware testing to evaluate dynamic behavior, mobile kinematics, control response, and sensor/actuator integration (SIGRobotics Website)

Texas Instruments | *Equipment Engineering Intern* *Lehi, UT* | May 2026 - August 2026

- Reduced ULK chamber downtime by 80% by designing and deploying a custom end-effector and automated ICC chamber clean routine, eliminating manual scrubbing and improving maintenance reliability
- Established an in-house refurbishment process for AMAT Producer showerheads, developing custom tooling jigs and validating cleaning procedures; filed 2 patent applications on the tooling IP and reduced replacement costs by >$15K per unit
- Assembled and sourced a clean-air purge system to mitigate hazardous HF and HCN outgassing in LAM Bevel chambers, improving equipment reliability and maintaining technician safety
- Developed custom Python scripts and user-friendly interfaces to mitigate data scraping bottlenecks and aid in figure construction, supporting equipment testing and failure analysis trends over time

Illini Electric Motorsports (FSAE) | *Team Member* *Urbana, IL* | August 2024 - December 2025

- Designed and iterated electric race-car brake system hardware in PTC Creo, including pedals, master-cylinder interfaces, calipers, pressure sensors, and brake-line routing for reliable operation and component compatibility
- Applied mechanical design and design-for-manufacturing principles to prototype and integrate brake components using machine-shop and electronics-lab resources
- Collaborated across mechanical, electrical, and software teams to troubleshoot subsystem interfaces, validate component fit and function, and integrate hardware into the vehicle
- Documented design iterations, system integration, material selection, cost analysis, testing, and troubleshooting procedures to support manufacturing and future development

TARC American Rocketry Challenge | *Team Lead, 2023 National Finalist* *Naperville, IL* | June 2022 - March 2024

- Led design, fabrication, and testing of a competition rocket optimized for altitude, flight duration, and payload safety, reaching the national finals from 800+ teams
- Conducted aerodynamic and structural analysis using OpenRocket, RockSim, and SolidWorks to iterate hardware designs and improve predictive performance

SKILLS

CAD & Manufacturing: Fusion 360, PTC Creo, Onshape, Autodesk Inventor, 3D Printing, Analysis, Machining, DFM, DFA
Robotics & Hardware: Prototyping, Electromechanical Integration, Hardware Testing, Sensors & Actuators, Failure Analysis, Troubleshooting, Tolerance Analysis, Data Acquisition, Design of Experiments
Programming & Simulation: Python, C++, Java, MATLAB, Git/GitHub, MuJoCo